SWCON221 Microservice Programming

Lecture 7

Docker Hub & Dockerfile

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Contents

- Docker hub
- Dockerfiles
- Lifecycle of a container
- ✴ Docker {Basic, Information, Clean-up} commands

Reading Articles

- Docker Hub

  https://docs.docker.com/docker-hub/

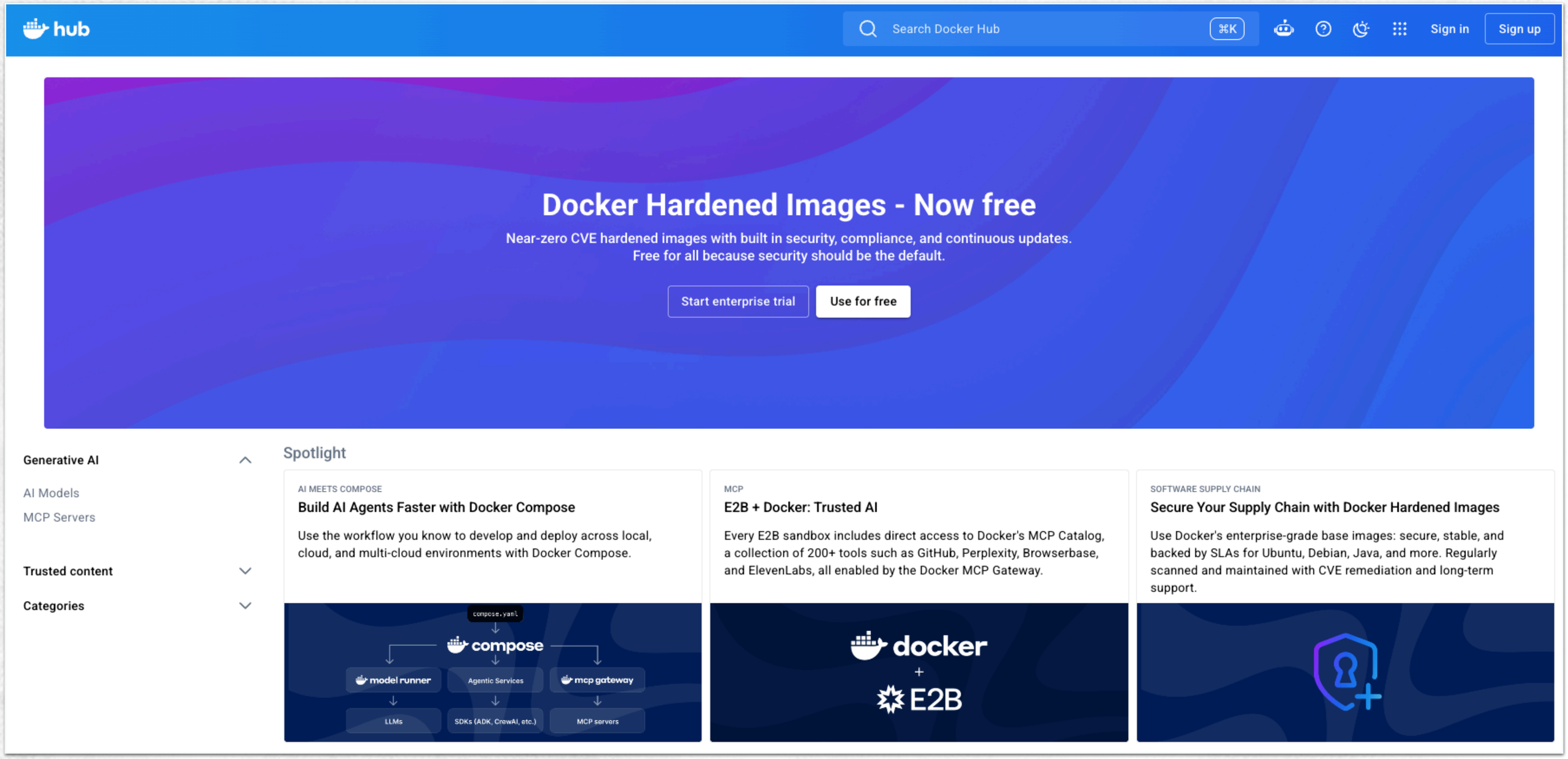
- Best practices for writing Dockerfiles

  https://docs.docker.com/develop/develop-images/dockerfile_best-practices/

FROM Lecture.5

Sharing Immutable Infrastructures

- Docker Hub (https://hub.docker.com/)



FROM Lecture.5

Sharing Immutable Infrastructures

- Docker Hub (https://hub.docker.com/)

IMAGE + 1 MORE

busybox

Docker Official Image

Busybox base image.

2m 1B+ 3.5K

IMAGE

alpine

Docker Official Image

A minimal Docker image based on Alpine Linux with a complete package index and only 5 MB in size!

25d 1B+ 10K+

IMAGE + 1 MORE

ubuntu

Docker Official Image

Ubuntu is a Debian-based Linux operating system based on free software.

1d 1B+ 10K+

IMAGE + 1 MORE

python

Docker Official Image

Python is an interpreted, interactive, object-oriented, open-source programming language.

1h 1B+ 10K+

IMAGE + 1 MORE

node

Docker Official Image

Node.js is a JavaScript-based platform for server-side and networking applications.

3d 1B+ 10K+

IMAGE + 1 MORE

golang

Docker Official Image

Go (golang) is a general purpose, higher-level, imperative programming language.

2d 1B+ 5.1K

MODEL + 1 MORE

ai/gemma4

Docker

Gemma 4: multimodal open AI models by Google, optimized for reasoning, coding, and long context.

1m 500K+ 41

MODEL

ai/nomic-embed-text-v1.5

Docker

Nomic Embed Text v1 is an open-source, fully auditable text embedding model

2m 10K+ 4

MODEL

ai/nemotron3

Docker

Multimodal LLM with video, audio, image, and text understanding for enterprise applications

3m 10K+ 3

IMAGE + 1 MORE

nginx

Docker Official Image

Official build of Nginx.

1d 1B+ 10K+

IMAGE + 1 MORE

httpd

Docker Official Image

The Apache HTTP Server Project

4d 1B+ 4.9K

IMAGE + 1 MORE

haproxy

Docker Official Image

HAProxy - The Reliable, High Performance TCP/HTTP Load Balancer

4d 1B+ 2.0K

Docker Hub Features

Generative AI

AI Models

MCP Servers

Trusted content

Docker Hardened Images

Docker Official Images

Verified Publisher

Sponsored OSS

Categories

Networking

Security

Languages & frameworks

Integration & delivery

Message queues

API management

Internet of things

Machine learning & AI

Developer tools

Data science

Web servers

Operating systems

Content management system

Databases & storage

Monitoring & observability

Web analytics

Reference: https://hub.docker.com/

Dockerfile

- A text file that contains all commands, in order, needed to build a given image
- Docker builds images automatically by reading the instructions from a Dockerfile
- A Docker image consists of read-only layers each of which represents a Dockerfile instruction.

Reference: https://docs.docker.com/develop/develop-images/dockerfile_best-practices/

Dockerfile Example

Install Python3 over Ubuntu

```
# syntax=docker/dockerfile:1
FROM ubuntu:24.04
RUN apt-get -y update && apt-get install -y --no-install-recommends python3
```

Execute .py using Python 3

```
FROM python:3

WORKDIR /usr/src/app

COPY requirements.txt ./
RUN pip install --no-cache-dir -r requirements.txt

COPY . .

CMD [ "python", "./your-daemon-or-script.py" ]
```


Reference: https://docs.docker.com/build/building/best-practices/
Reference: https://hub.docker.com/_/python

Dockerfile Example

- FROM: Specifies the base image that serves as the foundation for the container.
- WORKDIR: Sets the default working directory inside the container where commands will be executed.
- COPY: Copies files or directories from the host machine into the container.
- RUN: Executes commands, such as installing packages, during the image build process.
- CMD: Specifies the default command to run when the container starts.

Reference: https://docs.docker.com/build/building/best-practices/
Reference: https://hub.docker.com/_/python

🌟HOMEWORK🌟

Test-Drive: Codes

TESTDRIVE — root@2a7dffdf9209: / — -zsh — 80×11

```
drsungwon@/Users/drsungwon/Downloads/TESTDRIVE/:
$ more Dockerfile
FROM python:3-slim
WORKDIR /app
COPY . /app
CMD [ "python", "myapp.py" ]
drsungwon@/Users/drsungwon/Downloads/TESTDRIVE/:
$ more myapp.py
print("Hello, World!")
drsungwon@/Users/drsungwon/Downloads/TESTDRIVE/:
$
```

🌟HOMEWORK🌟

Test-Drive: Execution of Dockerfile

docker build -t myapp .
docker run -it --rm --name py3 myapp

```
TESTDRIVE — root@2a7dffdf9209: / — -zsh — 109×37
drsungwon@/Users/drsungwon/Downloads/TESTDRIVE/:
$ docker build -t myapp .

[+] Building 5.3s (8/8) FINISHED                                docker:desktop-linux
 => [internal] load build definition from Dockerfile                            0.0s
 => => transferring dockerfile: 109B                                            0.0s
 => [internal] load metadata for docker.io/library/python:3-slim                0.8s
 => [internal] load .dockerignore                                               0.0s
 => => transferring context: 99B                                                0.0s
 => [1/3] FROM docker.io/library/python:3-slim@sha256:cea0e6040540fb2b965       4.3s
 => => resolve docker.io/library/python:3-slim@sha256:cea0e6040540fb2b965       0.0s
 => => sha256:cea0e6040540fb2b965b6e7fb5ffa00871e632eef 10.37kB / 10.37kB       0.0s
 => => sha256:2f46bc1bd172fd75d27c6742da77deb977d91e9975a 1.75kB / 1.75kB       0.0s
 => => sha256:c42769af123efbff270d59f583b15e6c6c35829ad19 4.95kB / 4.95kB       0.0s
 => => sha256:59f54fbcd984beca03fd8b78569fa57268ecf78d2 30.14MB / 30.14MB       2.9s
 => => sha256:19873ad5d03358f70dc3d26ae8681995556c9460236 1.27MB / 1.27MB       1.0s
 => => sha256:db4514f465560e8d05f29c11f802ef829ec918c7b 12.25MB / 12.25MB       2.2s
 => => sha256:c879017f1fa945a043946db5f45d5ec18a205a5e6fa076a 249B / 249B       1.5s
 => => extracting sha256:59f54fbcd984beca03fd8b78569fa57268ecf78d291c0b6f       0.8s
 => => extracting sha256:19873ad5d03358f70dc3d26ae8681995556c9460236f0428       0.1s
 => => extracting sha256:db4514f465560e8d05f29c11f802ef829ec918c7bb405fe7       0.4s
 => => extracting sha256:c879017f1fa945a043946db5f45d5ec18a205a5e6fa076a8       0.0s
 => [internal] load build context                                               0.0s
 => => transferring context: 57B                                                0.0s
 => [2/3] WORKDIR /app                                                          0.3s
 => [3/3] COPY . /app                                                           0.0s
 => exporting to image                                                          0.0s
 => => exporting layers                                                         0.0s
 => => writing image sha256:a791cf39770731733ae4897c406cb58489252b445e474       0.0s
 => => naming to docker.io/library/myapp                                        0.0s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/wnna8ks87h0ej067f61tx21fr
drsungwon@/Users/drsungwon/Downloads/TESTDRIVE/:
$ docker run -it --rm --name py3 myapp
Hello, World!
drsungwon@/Users/drsungwon/Downloads/TESTDRIVE/:
$
```

Dockerfile Layers



Web Application

Layer 9 : CMD *Start Pintail.ai website*
Layer 8 : RUN *Commands to set up Pintail.ai Website*
Layer 7 : ADD *Pintail.ai binaries*
Layer 6 : FROM *Node.js - Apline Linux*

Application Framework

Layer 5 : RUN *Commands to set up Node.js*
Layer 4 : ADD *Node.js binaries*
Layer 3 : FROM *Alpine Linux*

Operating System

Layer 2 : RUN *Commands to set up OS*
Layer 1 : ADD *Operating System binaries*
Layer 0 : FROM *Scratch*

Reference: https://medium.com/pintail-labs/docker-series-creating-your-first-dockerfile-573bfea4991

Course

Dockerfile Goods

DOCKERFILE CHEAT SHEET - WHITE CERAMIC MUG

Size - 11oz

Quantity - 1

~~$13.99~~ *now* $11.99

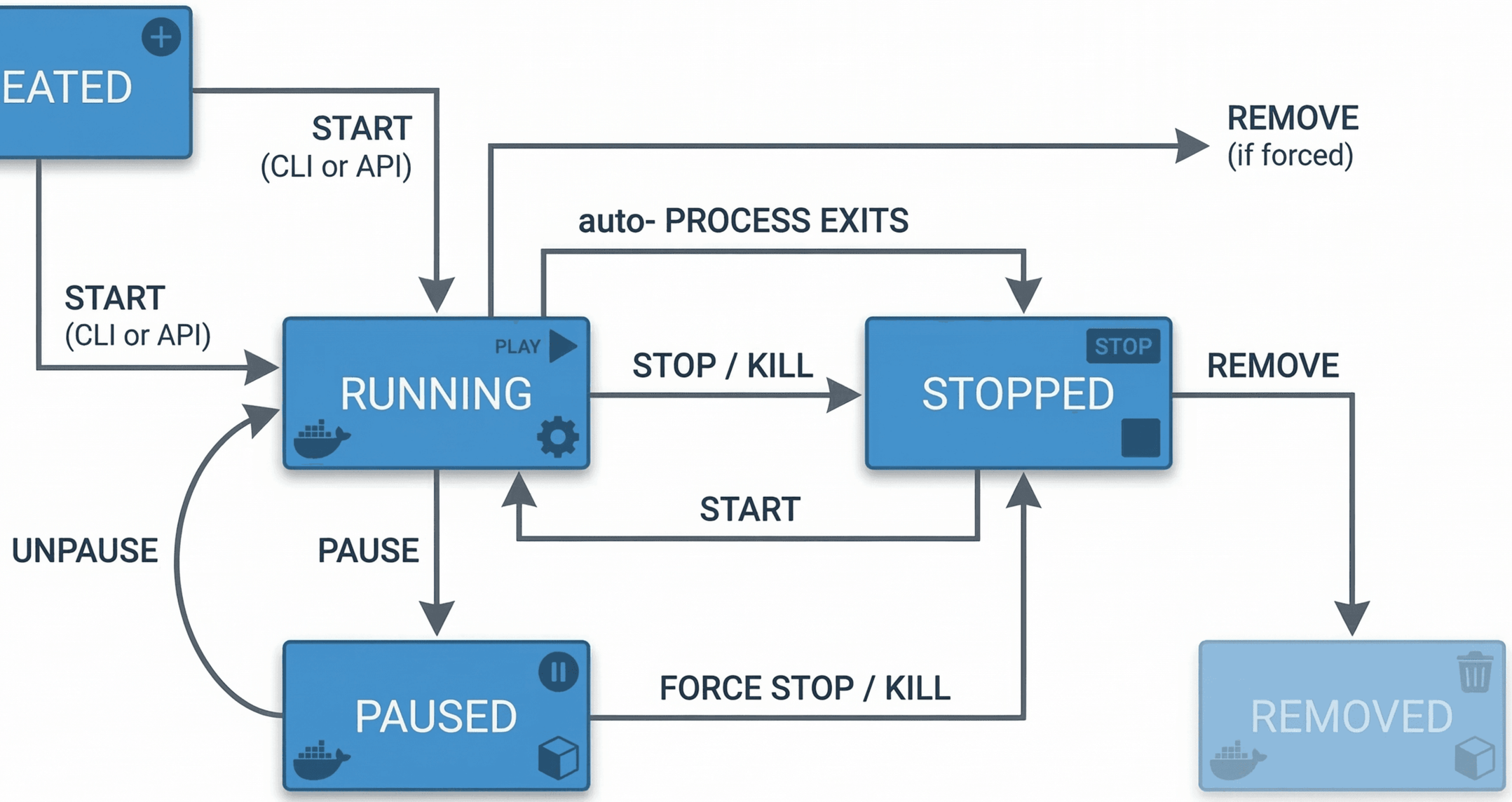
Tax included.

I WANT THIS! ADD TO CART.



Reference: https://remembertheapi.com/products/dockerfile-cheat-sheet-white-ceramic-mug

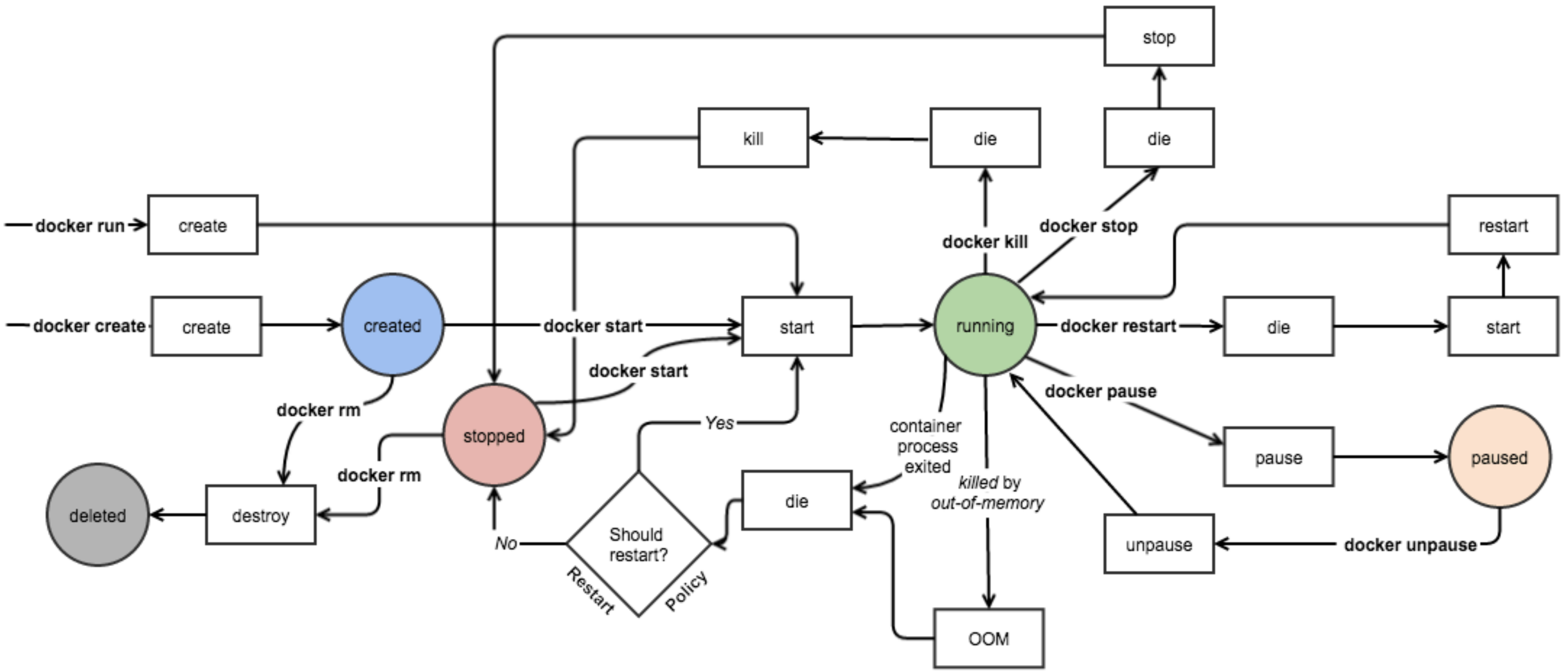
Lifecycle of Container



Reference: https://www.codingshuttle.com/blogs/docker-in-one-shot-the-ultimate-cheatsheet/

Lifecycle of Container (in detail)

Lifecycle of container with useful `docker container` commands



Reference: http://docker-saigon.github.io/post/Docker-Internals/

Basic Command

docker pull

docker build

docker image ls

docker container run

docker container ls

docker container ls -a (or --all)

docker container ls -a -q (or -aq)

docker container stop

docker container kill

Information Command

docker --version

docker system info

docker system df

docker ps

docker stats

docker search

Clean-up Command

docker image rm -f

docker container rm -f

docker system prune

Thank you

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